

Manual de Uso y Referencia Detallada de Funciones

1. Alcance

Este manual incluye uso funcional por rol y catalogo tecnico funcion por funcion del backend principal.

2. Roles del sistema

- * Usuario: crea, consulta, comenta, califica y reabre tickets dentro de la ventana permitida.
- * Tecnico: gestiona flujo operativo de tickets, chat y conocimientos.
- * Administrador: gestiona configuracion, seguridad, usuarios, integraciones, auditoria y reportes.

3. Modulos principales

- * app/auth.py: autenticacion, password, 2FA y cierre de sesion.
- * app/tickets.py: operaciones completas de tickets, chat y reaperturas.
- * app/admin.py: panel administrativo, settings, KB, integraciones y auditoria.
- * app/api.py: endpoints externos y webhooks.
- * app/routes.py: paginas publicas, health y utilidades de sesion.
- * app/models.py: entidades y reglas de negocio de datos.

4. Referencia detallada de funciones

La siguiente seccion describe cada funcion detectada en el paquete app.

Modulo: app/__init__.py

Funcion: utcnow

- * Ubicacion: app/__init__.py:44
- * Tipo: Application helper
- * Parametros: sin parametros declarados
- * Que hace: Provides shared application utility behavior used by multiple flows.

Funcion: _rate_limit_key

- * Ubicacion: app/__init__.py:49
- * Tipo: Internal helper
- * Parametros: sin parametros declarados
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.

Funcion: translate

- * Ubicacion: app/__init__.py:894
- * Tipo: Application helper
- * Parametros: text, lang
- * Que hace: Provides shared application utility behavior used by multiple flows.

Funcion: send_email

- * Ubicacion: app/__init__.py:898
- * Tipo: Application helper
- * Parametros: subject, recipients, body, html
- * Que hace: Provides shared application utility behavior used by multiple flows.
- * Efectos principales:
- * Triggers email notification flow.
- * Triggers webhook notification flow.

Funcion: send_webhook

- * Ubicacion: app/__init__.py:913
- * Tipo: Application helper
- * Parametros: message
- * Que hace: Provides shared application utility behavior used by multiple flows.
- * Efectos principales:
- * Triggers webhook notification flow.

Funcion: _validate_secret_key

- * Ubicacion: app/__init__.py:926
- * Tipo: Internal helper
- * Parametros: app
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.

Funcion: create_app

- * Ubicacion: app/__init__.py:953
- * Tipo: Application helper
- * Parametros: config_class
- * Que hace: Provides shared application utility behavior used by multiple flows.
- * Efectos principales:
- * Persists database changes.
- * Creates or stages new records in database session.
- * Rolls back transaction on error.
- * Renders HTML template response.

- * Redirects browser to another route.

Funcion: load_user

- * Ubicacion: app/__init__.py:1046
- * Tipo: Application helper
- * Parametros: user_id
- * Que hace: Provides shared application utility behavior used by multiple flows.

Funcion: inject_current_year

- * Ubicacion: app/__init__.py:1072
- * Tipo: Application helper
- * Parametros: sin parametros declarados
- * Que hace: Provides shared application utility behavior used by multiple flows.

Funcion: _translate

- * Ubicacion: app/__init__.py:1075
- * Tipo: Internal helper
- * Parametros: text
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.

Funcion: from_json_filter

- * Ubicacion: app/__init__.py:1086
- * Tipo: Application helper
- * Parametros: value
- * Que hace: Provides shared application utility behavior used by multiple flows.

Funcion: update_user_activity_and_enforce_password

- * Ubicacion: app/__init__.py:1097
- * Tipo: Application helper
- * Parametros: sin parametros declarados
- * Que hace: Provides shared application utility behavior used by multiple flows.
- * Efectos principales:
- * Persists database changes.
- * Rolls back transaction on error.
- * Redirects browser to another route.

Funcion: handle_unexpected_error

- * Ubicacion: app/__init__.py:1156

- * Tipo: Application helper
- * Parametros: error
- * Que hace: Provides shared application utility behavior used by multiple flows.
- * Efectos principales:
- * Renders HTML template response.

Modulo: app/admin.py

Funcion: utcnow

- * Ubicacion: app/admin.py:18
- * Tipo: Application helper
- * Parametros: sin parametros declarados
- * Que hace: Provides shared application utility behavior used by multiple flows.

Funcion: _t

- * Ubicacion: app/admin.py:23
- * Tipo: Internal helper
- * Parametros: text
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.

Funcion: admin_required

- * Ubicacion: app/admin.py:27
- * Tipo: Application helper
- * Parametros: func
- * Que hace: Provides shared application utility behavior used by multiple flows.
- * Efectos principales:
- * Sends user-visible feedback message.
- * Redirects browser to another route.

Funcion: wrapper

- * Ubicacion: app/admin.py:32
- * Tipo: Application helper
- * Parametros: sin parametros declarados
- * Que hace: Provides shared application utility behavior used by multiple flows.
- * Efectos principales:
- * Sends user-visible feedback message.
- * Redirects browser to another route.

Funcion: tech_or_admin_required

- * Ubicacion: app/admin.py:41
- * Tipo: Application helper
- * Parametros: func
- * Que hace: Provides shared application utility behavior used by multiple flows.
- * Efectos principales:
- * Sends user-visible feedback message.
- * Redirects browser to another route.

Funcion: wrapper

- * Ubicacion: app/admin.py:46
- * Tipo: Application helper
- * Parametros: sin parametros declarados
- * Que hace: Provides shared application utility behavior used by multiple flows.
- * Efectos principales:
- * Sends user-visible feedback message.
- * Redirects browser to another route.

Funcion: list_users

- * Ubicacion: app/admin.py:58
- * Tipo: HTTP endpoint
- * Ruta: /users
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /users with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:
- * Rolls back transaction on error.
- * Renders HTML template response.
- * Sends user-visible feedback message.

Funcion: create_user

- * Ubicacion: app/admin.py:73
- * Tipo: HTTP endpoint
- * Ruta: /users/create

- * Metodos HTTP: GET, POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /users/create with HTTP methods GET, POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:
- * Persists database changes.
- * Creates or stages new records in database session.
- * Rolls back transaction on error.
- * Renders HTML template response.
- * Sends user-visible feedback message.
- * Redirects browser to another route.

Funcion: edit_user

- * Ubicacion: app/admin.py:102
- * Tipo: HTTP endpoint
- * Ruta: /users/edit
- * Metodos HTTP: GET, POST
- * Parametros: user_id
- * Que hace: Exposes route /users/edit with HTTP methods GET, POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:
- * Persists database changes.
- * Renders HTML template response.
- * Sends user-visible feedback message.
- * Loads resource or returns 404 automatically.
- * Redirects browser to another route.

Funcion: reset_user_password

- * Ubicacion: app/admin.py:118
- * Tipo: HTTP endpoint
- * Ruta: /users/reset-password
- * Metodos HTTP: GET, POST

- * Parametros: user_id
- * Que hace: Exposes route /users//reset-password with HTTP methods GET, POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:
- * Persists database changes.
- * Rolls back transaction on error.
- * Renders HTML template response.
- * Sends user-visible feedback message.
- * Loads resource or returns 404 automatically.
- * Redirects browser to another route.

Funcion: dashboard

- * Ubicacion: app/admin.py:173
- * Tipo: HTTP endpoint
- * Ruta: /dashboard
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /dashboard with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires technician or administrator role.
- * Efectos principales:
- * Renders HTML template response.

Funcion: chat_monitoring

- * Ubicacion: app/admin.py:258
- * Tipo: HTTP endpoint
- * Ruta: /chat-monitoring
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /chat-monitoring with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.

- * Efectos principales:
- * Renders HTML template response.
- * Sends user-visible feedback message.
- * Redirects browser to another route.

Funcion: _get_chat_participants

- * Ubicacion: app/admin.py:299
- * Tipo: Internal helper
- * Parametros: ticket
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.

Funcion: online_users_list

- * Ubicacion: app/admin.py:334
- * Tipo: HTTP endpoint
- * Ruta: /online-users
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /online-users with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:
- * Renders HTML template response.

Funcion: online_users_data

- * Ubicacion: app/admin.py:361
- * Tipo: HTTP endpoint
- * Ruta: /online-users/data
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /online-users/data with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.

Funcion: debug_online_users_json

- * Ubicacion: app/admin.py:390

- * Tipo: HTTP endpoint
- * Ruta: /debug/online-users
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /debug/online-users with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.

Funcion: ticket_options

- * Ubicacion: app/admin.py:416
- * Tipo: HTTP endpoint
- * Ruta: /ticket-options
- * Metodos HTTP: GET, POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /ticket-options with HTTP methods GET, POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:
- * Persists database changes.
- * Creates or stages new records in database session.
- * Renders HTML template response.
- * Sends user-visible feedback message.
- * Redirects browser to another route.

Funcion: delete_ticket_option

- * Ubicacion: app/admin.py:441
- * Tipo: HTTP endpoint
- * Ruta: /ticket-options//delete
- * Metodos HTTP: POST
- * Parametros: option_id
- * Que hace: Exposes route /ticket-options//delete with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.

- * Efectos principales:
- * Persists database changes.
- * Sends user-visible feedback message.
- * Loads resource or returns 404 automatically.
- * Redirects browser to another route.

Funcion: edit_ticket_option

- * Ubicacion: app/admin.py:452
- * Tipo: HTTP endpoint
- * Ruta: /ticket-options//edit
- * Metodos HTTP: POST
- * Parametros: option_id
- * Que hace: Exposes route /ticket-options//edit with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:
- * Persists database changes.
- * Sends user-visible feedback message.
- * Loads resource or returns 404 automatically.
- * Redirects browser to another route.

Funcion: ml_system

- * Ubicacion: app/admin.py:478
- * Tipo: HTTP endpoint
- * Ruta: /ml-system
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /ml-system with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:
- * Renders HTML template response.
- * Sends user-visible feedback message.
- * Redirects browser to another route.

Funcion: ml_train

- * Ubicacion: app/admin.py:535
- * Tipo: HTTP endpoint
- * Ruta: /ml-system/train
- * Metodos HTTP: POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /ml-system/train with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
 - * Requires authenticated session.
 - * Requires administrator privileges.
- * Efectos principales:
 - * Sends user-visible feedback message.
 - * Redirects browser to another route.

Funcion: integrations

- * Ubicacion: app/admin.py:552
- * Tipo: HTTP endpoint
- * Ruta: /integrations
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /integrations with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
 - * Requires authenticated session.
 - * Requires administrator privileges.
- * Efectos principales:
 - * Redirects browser to another route.

Funcion: create_api_token

- * Ubicacion: app/admin.py:560
- * Tipo: HTTP endpoint
- * Ruta: /integrations/tokens/create
- * Metodos HTTP: POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /integrations/tokens/create with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
 - * Requires authenticated session.

- * Requires administrator privileges.
- * Efectos principales:
- * Persists database changes.
- * Creates or stages new records in database session.
- * Sends user-visible feedback message.
- * Redirects browser to another route.

Funcion: revoke_api_token

- * Ubicacion: app/admin.py:591
- * Tipo: HTTP endpoint
- * Ruta: /integrations/tokens//revoke
- * Metodos HTTP: POST
- * Parametros: token_id
- * Que hace: Exposes route /integrations/tokens//revoke with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:
- * Persists database changes.
- * Sends user-visible feedback message.
- * Loads resource or returns 404 automatically.
- * Redirects browser to another route.

Funcion: create_integration

- * Ubicacion: app/admin.py:604
- * Tipo: HTTP endpoint
- * Ruta: /integrations/create
- * Metodos HTTP: POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /integrations/create with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:
- * Persists database changes.
- * Creates or stages new records in database session.

- * Sends user-visible feedback message.
- * Redirects browser to another route.

Funcion: toggle_integration

- * Ubicacion: app/admin.py:641
- * Tipo: HTTP endpoint
- * Ruta: /integrations//toggle
- * Metodos HTTP: POST
- * Parametros: integration_id
- * Que hace: Exposes route /integrations//toggle with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
 - * Requires authenticated session.
 - * Requires administrator privileges.
- * Efectos principales:
 - * Persists database changes.
 - * Sends user-visible feedback message.
 - * Loads resource or returns 404 automatically.
 - * Redirects browser to another route.

Funcion: delete_integration

- * Ubicacion: app/admin.py:655
- * Tipo: HTTP endpoint
- * Ruta: /integrations//delete
- * Metodos HTTP: POST
- * Parametros: integration_id
- * Que hace: Exposes route /integrations//delete with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
 - * Requires authenticated session.
 - * Requires administrator privileges.
- * Efectos principales:
 - * Persists database changes.
 - * Sends user-visible feedback message.
 - * Loads resource or returns 404 automatically.
 - * Redirects browser to another route.

Funcion: ai_system

- * Ubicacion: app/admin.py:668

- * Tipo: HTTP endpoint
- * Ruta: /ai-system
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /ai-system with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:
- * Renders HTML template response.

Funcion: ai_system_settings

- * Ubicacion: app/admin.py:692
- * Tipo: HTTP endpoint
- * Ruta: /ai-system/settings
- * Metodos HTTP: POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /ai-system/settings with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:
- * Sends user-visible feedback message.
- * Redirects browser to another route.

Funcion: settings

- * Ubicacion: app/admin.py:714
- * Tipo: HTTP endpoint
- * Ruta: /settings
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /settings with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:

- * Renders HTML template response.

Funcion: settings_users

- * Ubicacion: app/admin.py:722
- * Tipo: HTTP endpoint
- * Ruta: /settings/users
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /settings/users with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
 - * Requires authenticated session.
 - * Requires administrator privileges.
- * Efectos principales:
 - * Rolls back transaction on error.
 - * Renders HTML template response.
 - * Sends user-visible feedback message.

Funcion: settings_ticket_options

- * Ubicacion: app/admin.py:743
- * Tipo: HTTP endpoint
- * Ruta: /settings/ticket-options
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /settings/ticket-options with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
 - * Requires authenticated session.
 - * Requires administrator privileges.
- * Efectos principales:
 - * Persists database changes.
 - * Creates or stages new records in database session.
 - * Rolls back transaction on error.
 - * Renders HTML template response.
 - * Sends user-visible feedback message.
 - * Redirects browser to another route.

Funcion: settings_ml

- * Ubicacion: app/admin.py:782

- * Tipo: HTTP endpoint
- * Ruta: /settings/ml
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /settings/ml with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:
- * Renders HTML template response.

Funcion: settings_ai

- * Ubicacion: app/admin.py:829
- * Tipo: HTTP endpoint
- * Ruta: /settings/ai
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /settings/ai with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:
- * Renders HTML template response.

Funcion: settings_ai_update

- * Ubicacion: app/admin.py:853
- * Tipo: HTTP endpoint
- * Ruta: /settings/ai/settings
- * Metodos HTTP: POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /settings/ai/settings with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:
- * Sends user-visible feedback message.

- * Redirects browser to another route.

Funcion: settings_integrations

- * Ubicacion: app/admin.py:869
- * Tipo: HTTP endpoint
- * Ruta: /settings/integrations
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /settings/integrations with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
 - * Requires authenticated session.
 - * Requires administrator privileges.
- * Efectos principales:
- * Renders HTML template response.

Funcion: settings_audit

- * Ubicacion: app/admin.py:897
- * Tipo: HTTP endpoint
- * Ruta: /settings/audit
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /settings/audit with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
 - * Requires authenticated session.
 - * Requires administrator privileges.
- * Efectos principales:
- * Renders HTML template response.

Funcion: settings_audit_export

- * Ubicacion: app/admin.py:918
- * Tipo: HTTP endpoint
- * Ruta: /settings/audit/export
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /settings/audit/export with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:

- * Requires authenticated session.
- * Requires administrator privileges.

Funcion: settings_office365_email_update

- * Ubicacion: app/admin.py:959
- * Tipo: HTTP endpoint
- * Ruta: /settings/integrations/office365-email
- * Metodos HTTP: POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /settings/integrations/office365-email with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
 - * Requires authenticated session.
 - * Requires administrator privileges.
- * Efectos principales:
 - * Persists database changes.
 - * Creates or stages new records in database session.
 - * Sends user-visible feedback message.
 - * Redirects browser to another route.

Funcion: kb_list

- * Ubicacion: app/admin.py:996
- * Tipo: HTTP endpoint
- * Ruta: /kb
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /kb with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
 - * Requires authenticated session.
 - * Requires technician or administrator role.
- * Efectos principales:
 - * Renders HTML template response.

Funcion: kb_search

- * Ubicacion: app/admin.py:1021
- * Tipo: HTTP endpoint
- * Ruta: /kb/search
- * Metodos HTTP: GET

- * Parametros: sin parametros declarados
- * Que hace: Exposes route /kb/search with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires technician or administrator role.
- * Efectos principales:
- * Returns JSON response.

Funcion: kb_from_comment

- * Ubicacion: app/admin.py:1042
- * Tipo: HTTP endpoint
- * Ruta: /kb/from-comment/
- * Metodos HTTP: GET
- * Parametros: comment_id
- * Que hace: Exposes route /kb/from-comment/ with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires technician or administrator role.
- * Efectos principales:
- * Loads resource or returns 404 automatically.
- * Redirects browser to another route.

Funcion: kb_from_ticket

- * Ubicacion: app/admin.py:1057
- * Tipo: HTTP endpoint
- * Ruta: /kb/from-ticket/
- * Metodos HTTP: GET
- * Parametros: ticket_id
- * Que hace: Exposes route /kb/from-ticket/ with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires technician or administrator role.
- * Efectos principales:
- * Sends user-visible feedback message.
- * Loads resource or returns 404 automatically.
- * Redirects browser to another route.

Funcion: kb_create

- * Ubicacion: app/admin.py:1082
- * Tipo: HTTP endpoint
- * Ruta: /kb/create
- * Metodos HTTP: GET, POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /kb/create with HTTP methods GET, POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
 - * Requires authenticated session.
 - * Requires technician or administrator role.
- * Efectos principales:
 - * Persists database changes.
 - * Creates or stages new records in database session.
 - * Renders HTML template response.
 - * Sends user-visible feedback message.
 - * Redirects browser to another route.

Funcion: kb_edit

- * Ubicacion: app/admin.py:1118
- * Tipo: HTTP endpoint
- * Ruta: /kb/edit
- * Metodos HTTP: GET, POST
- * Parametros: article_id
- * Que hace: Exposes route /kb/edit with HTTP methods GET, POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
 - * Requires authenticated session.
 - * Requires technician or administrator role.
- * Efectos principales:
 - * Persists database changes.
 - * Renders HTML template response.
 - * Sends user-visible feedback message.
 - * Loads resource or returns 404 automatically.
 - * Redirects browser to another route.

Funcion: kb_delete

- * Ubicacion: app/admin.py:1142

- * Tipo: HTTP endpoint
- * Ruta: /kb//delete
- * Metodos HTTP: POST
- * Parametros: article_id
- * Que hace: Exposes route /kb//delete with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:
- * Persists database changes.
- * Sends user-visible feedback message.
- * Loads resource or returns 404 automatically.
- * Redirects browser to another route.

Funcion: kb_approve

- * Ubicacion: app/admin.py:1154
- * Tipo: HTTP endpoint
- * Ruta: /kb//approve
- * Metodos HTTP: POST
- * Parametros: article_id
- * Que hace: Exposes route /kb//approve with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:
- * Persists database changes.
- * Sends user-visible feedback message.
- * Loads resource or returns 404 automatically.
- * Redirects browser to another route.

Funcion: kb_reject

- * Ubicacion: app/admin.py:1168
- * Tipo: HTTP endpoint
- * Ruta: /kb//reject
- * Metodos HTTP: POST
- * Parametros: article_id

- * Que hace: Exposes route /kb//reject with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires administrator privileges.
- * Efectos principales:
- * Persists database changes.
- * Creates or stages new records in database session.
- * Sends user-visible feedback message.
- * Loads resource or returns 404 automatically.
- * Redirects browser to another route.

Funcion: executive_report

- * Ubicacion: app/admin.py:1196
- * Tipo: HTTP endpoint
- * Ruta: /report
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /report with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Requires technician or administrator role.
- * Efectos principales:
- * Renders HTML template response.

Modulo: app/ai_chatbot.py

Funcion: ChatBot.__init__

- * Ubicacion: app/ai_chatbot.py:178
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: ChatBot._train

- * Ubicacion: app/ai_chatbot.py:185
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: ChatBot.find_answer

- * Ubicacion: app/ai_chatbot.py:204
- * Tipo: Class method
- * Parametros: self, question
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: ChatBot.get_all_categories

- * Ubicacion: app/ai_chatbot.py:253
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: ChatBot.get_entries_by_category

- * Ubicacion: app/ai_chatbot.py:257
- * Tipo: Class method
- * Parametros: self, category
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: get_chatbot

- * Ubicacion: app/ai_chatbot.py:266
- * Tipo: Application helper
- * Parametros: sin parametros declarados
- * Que hace: Provides shared application utility behavior used by multiple flows.

Funcion: answer_question

- * Ubicacion: app/ai_chatbot.py:274
- * Tipo: Application helper
- * Parametros: question
- * Que hace: Provides shared application utility behavior used by multiple flows.

Funcion: should_auto_respond

- * Ubicacion: app/ai_chatbot.py:287
- * Tipo: Application helper
- * Parametros: title, description, confidence_threshold
- * Que hace: Provides shared application utility behavior used by multiple flows.

Modulo: app/ai_sentiment.py

Funcion: SentimentAnalyzer.__init__

- * Ubicacion: app/ai_sentiment.py:38
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: SentimentAnalyzer.analyze

- * Ubicacion: app/ai_sentiment.py:44
- * Tipo: Class method
- * Parametros: self, text
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: SentimentAnalyzer._get_textblob_sentiment

- * Ubicacion: app/ai_sentiment.py:90
- * Tipo: Class method
- * Parametros: self, text
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: SentimentAnalyzer._get_keyword_sentiment

- * Ubicacion: app/ai_sentiment.py:106
- * Tipo: Class method
- * Parametros: self, text_lower
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: SentimentAnalyzer._label_sentiment

- * Ubicacion: app/ai_sentiment.py:123
- * Tipo: Class method
- * Parametros: self, score
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: SentimentAnalyzer._analyze_urgency

- * Ubicacion: app/ai_sentiment.py:136
- * Tipo: Class method
- * Parametros: self, text_lower
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: SentimentAnalyzer._build_explanation

- * Ubicacion: app/ai_sentiment.py:165

- * Tipo: Class method
- * Parametros: self, sentiment, urgency, text_lower
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: SentimentAnalyzer._default_result

- * Ubicacion: app/ai_sentiment.py:189
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: SentimentAnalyzer.get_priority_adjustment

- * Ubicacion: app/ai_sentiment.py:200
- * Tipo: Class method
- * Parametros: self, sentiment_label
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: SentimentAnalyzer.get_urgency_adjustment

- * Ubicacion: app/ai_sentiment.py:218
- * Tipo: Class method
- * Parametros: self, urgency_level
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: get_analyzer

- * Ubicacion: app/ai_sentiment.py:239
- * Tipo: Application helper
- * Parametros: sin parametros declarados
- * Que hace: Provides shared application utility behavior used by multiple flows.

Funcion: analyze_ticket

- * Ubicacion: app/ai_sentiment.py:247
- * Tipo: Application helper
- * Parametros: title, description
- * Que hace: Provides shared application utility behavior used by multiple flows.

Modulo: app/api.py

Funcion: _integration_config

- * Ubicacion: app/api.py:20

- * Tipo: Internal helper
- * Parametros: integration
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.

Funcion: require_api_token

- * Ubicacion: app/api.py:29
- * Tipo: Application helper
- * Parametros: f
- * Que hace: Provides shared application utility behavior used by multiple flows.
- * Efectos principales:
- * Persists database changes.
- * Returns JSON response.

Funcion: decorated_function

- * Ubicacion: app/api.py:32
- * Tipo: Application helper
- * Parametros: sin parametros declarados
- * Que hace: Provides shared application utility behavior used by multiple flows.
- * Efectos principales:
- * Persists database changes.
- * Returns JSON response.

Funcion: create_ticket_api

- * Ubicacion: app/api.py:61
- * Tipo: HTTP endpoint
- * Ruta: /tickets
- * Metodos HTTP: POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /tickets with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Efectos principales:
- * Persists database changes.
- * Creates or stages new records in database session.
- * Rolls back transaction on error.
- * Returns JSON response.

Funcion: office365_email_intake

- * Ubicacion: app/api.py:210

- * Tipo: HTTP endpoint
- * Ruta: /email/intake
- * Metodos HTTP: POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /email/intake with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Efectos principales:
- * Persists database changes.
- * Creates or stages new records in database session.
- * Rolls back transaction on error.
- * Returns JSON response.

Funcion: get_ticket_api

- * Ubicacion: app/api.py:271
- * Tipo: HTTP endpoint
- * Ruta: /tickets/
- * Metodos HTTP: GET
- * Parametros: ticket_id
- * Que hace: Exposes route /tickets/ with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Efectos principales:
- * Returns JSON response.
- * Loads resource or returns 404 automatically.

Funcion: slack_webhook

- * Ubicacion: app/api.py:295
- * Tipo: HTTP endpoint
- * Ruta: /webhooks/slack
- * Metodos HTTP: POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /webhooks/slack with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Efectos principales:
- * Persists database changes.
- * Creates or stages new records in database session.
- * Returns JSON response.

Funcion: teams_webhook

- * Ubicacion: app/api.py:410

- * Tipo: HTTP endpoint
- * Ruta: /webhooks/teams
- * Metodos HTTP: POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /webhooks/teams with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Efectos principales:
- * Persists database changes.
- * Creates or stages new records in database session.
- * Returns JSON response.

Modulo: app/api_validators.py

Funcion: TicketCreateSchema.process_data

- * Ubicacion: app/api_validators.py:16
- * Tipo: Class method
- * Parametros: self, data
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: TicketUpdateSchema.validate_status

- * Ubicacion: app/api_validators.py:32
- * Tipo: Class method
- * Parametros: self, data
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: TicketCommentSchema.process_data

- * Ubicacion: app/api_validators.py:44
- * Tipo: Class method
- * Parametros: self, data
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: WebhookPayloadSchema.validate_timestamp

- * Ubicacion: app/api_validators.py:59
- * Tipo: Class method
- * Parametros: self, data
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: PasswordChangeSchema.validate_passwords_match

- * Ubicacion: app/api_validators.py:95
- * Tipo: Class method
- * Parametros: self, data
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: validate_request

- * Ubicacion: app/api_validators.py:104
- * Tipo: Application helper
- * Parametros: schema_class, data
- * Que hace: Provides shared application utility behavior used by multiple flows.

Modulo: app/auth.py

Funcion: load_user

- * Ubicacion: app/auth.py:14
- * Tipo: Application helper
- * Parametros: user_id
- * Que hace: Provides shared application utility behavior used by multiple flows.

Funcion: _t

- * Ubicacion: app/auth.py:20
- * Tipo: Internal helper
- * Parametros: text
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.

Funcion: must_change_default_admin_password

- * Ubicacion: app/auth.py:24
- * Tipo: Application helper
- * Parametros: user
- * Que hace: Provides shared application utility behavior used by multiple flows.

Funcion: _login_rate_limit_key

- * Ubicacion: app/auth.py:35
- * Tipo: Internal helper
- * Parametros: sin parametros declarados
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.

Funcion: login

- * Ubicacion: app/auth.py:45
- * Tipo: HTTP endpoint
- * Ruta: /login
- * Metodos HTTP: GET, POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /login with HTTP methods GET, POST. Handles web request lifecycle and returns UI/API output.
- * Efectos principales:
- * Renders HTML template response.
- * Sends user-visible feedback message.
- * Redirects browser to another route.

Funcion: force_password_change

- * Ubicacion: app/auth.py:99
- * Tipo: HTTP endpoint
- * Ruta: /force-password-change
- * Metodos HTTP: GET, POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /force-password-change with HTTP methods GET, POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Efectos principales:
- * Persists database changes.
- * Renders HTML template response.
- * Sends user-visible feedback message.
- * Redirects browser to another route.

Funcion: change_password

- * Ubicacion: app/auth.py:137
- * Tipo: HTTP endpoint
- * Ruta: /change-password
- * Metodos HTTP: GET, POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /change-password with HTTP methods GET, POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Efectos principales:

- * Persists database changes.
- * Renders HTML template response.
- * Sends user-visible feedback message.
- * Redirects browser to another route.

Funcion: verify_totp

- * Ubicacion: app/auth.py:183
- * Tipo: HTTP endpoint
- * Ruta: /verify-2fa
- * Metodos HTTP: GET, POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /verify-2fa with HTTP methods GET, POST. Handles web request lifecycle and returns UI/API output.
- * Efectos principales:
 - * Persists database changes.
 - * Renders HTML template response.
 - * Sends user-visible feedback message.
 - * Redirects browser to another route.

Funcion: logout

- * Ubicacion: app/auth.py:227
- * Tipo: HTTP endpoint
- * Ruta: /logout
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /logout with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Efectos principales:
 - * Sends user-visible feedback message.
 - * Redirects browser to another route.

Funcion: register

- * Ubicacion: app/auth.py:234
- * Tipo: HTTP endpoint
- * Ruta: /register
- * Metodos HTTP: GET, POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /register with HTTP methods GET, POST. Handles web request lifecycle and returns UI/API output.

- * Efectos principales:
- * Can terminate request with HTTP error code.

Modulo: app/forms.py

Funcion: tr

- * Ubicacion: app/forms.py:11
- * Tipo: Application helper
- * Parametros: en_text, es_text
- * Que hace: Provides shared application utility behavior used by multiple flows.

Funcion: LoginForm.__init__

- * Ubicacion: app/forms.py:21
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: TicketForm.__init__

- * Ubicacion: app/forms.py:38
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: TicketUpdateForm.__init__

- * Ubicacion: app/forms.py:64
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: CSATForm.__init__

- * Ubicacion: app/forms.py:87
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: UserRoleForm.__init__

- * Ubicacion: app/forms.py:98
- * Tipo: Class method

- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: company_email

- * Ubicacion: app/forms.py:105
- * Tipo: Application helper
- * Parametros: form, field
- * Que hace: Provides shared application utility behavior used by multiple flows.

Funcion: NewUserForm.__init__

- * Ubicacion: app/forms.py:122
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: NewUserForm.validate_username

- * Ubicacion: app/forms.py:131
- * Tipo: Class method
- * Parametros: self, field
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: NewUserForm.validate_email

- * Ubicacion: app/forms.py:136
- * Tipo: Class method
- * Parametros: self, field
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: TicketOptionForm.__init__

- * Ubicacion: app/forms.py:147
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: TicketOptionForm.validate_value

- * Ubicacion: app/forms.py:155
- * Tipo: Class method
- * Parametros: self, field
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: ForcePasswordChangeForm.__init__

- * Ubicacion: app/forms.py:166
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: AdminResetPasswordForm.__init__

- * Ubicacion: app/forms.py:178
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: ChangePasswordForm.__init__

- * Ubicacion: app/forms.py:191
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Modulo: app/ml_classifier.py

Funcion: TicketClassifier.__init__

- * Ubicacion: app/ml_classifier.py:35
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: TicketClassifier._prepare_features

- * Ubicacion: app/ml_classifier.py:48
- * Tipo: Class method
- * Parametros: self, tickets_data
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: TicketClassifier.train

- * Ubicacion: app/ml_classifier.py:89
- * Tipo: Class method
- * Parametros: self, min_tickets
- * Que hace: Implements model/business behavior encapsulated in class logic.
- * Efectos principales:

- * Persists database changes.

Funcion: TicketClassifier.predict

- * Ubicacion: app/ml_classifier.py:182
- * Tipo: Class method
- * Parametros: self, ticket_data, top_n
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: TicketClassifier._update_technician_stats

- * Ubicacion: app/ml_classifier.py:237
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.
- * Efectos principales:
- * Persists database changes.
- * Creates or stages new records in database session.

Funcion: TicketClassifier.save_model

- * Ubicacion: app/ml_classifier.py:288
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: TicketClassifier.load_model

- * Ubicacion: app/ml_classifier.py:309
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: TicketClassifier.get_model_info

- * Ubicacion: app/ml_classifier.py:332
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Modulo: app/ml_commands.py

Funcion: register_ml_commands

- * Ubicacion: app/ml_commands.py:10
- * Tipo: Application helper
- * Parametros: app
- * Que hace: Provides shared application utility behavior used by multiple flows.

Funcion: train_model

- * Ubicacion: app/ml_commands.py:15
- * Tipo: Application helper
- * Parametros: min_tickets
- * Que hace: Provides shared application utility behavior used by multiple flows.

Funcion: model_info

- * Ubicacion: app/ml_commands.py:42
- * Tipo: Application helper
- * Parametros: sin parametros declarados
- * Que hace: Provides shared application utility behavior used by multiple flows.

Funcion: technician_stats

- * Ubicacion: app/ml_commands.py:67
- * Tipo: Application helper
- * Parametros: sin parametros declarados
- * Que hace: Provides shared application utility behavior used by multiple flows.

Modulo: app/models.py

Funcion: utcnow

- * Ubicacion: app/models.py:11
- * Tipo: Application helper
- * Parametros: sin parametros declarados
- * Que hace: Provides shared application utility behavior used by multiple flows.

Funcion: User.set_password

- * Ubicacion: app/models.py:36
- * Tipo: Class method
- * Parametros: self, password
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: User.check_password

- * Ubicacion: app/models.py:40
- * Tipo: Class method
- * Parametros: self, password
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: User.is_admin

- * Ubicacion: app/models.py:43
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: User.is_technician

- * Ubicacion: app/models.py:46
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: User.is_account_locked

- * Ubicacion: app/models.py:49
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.
- * Efectos principales:
- * Persists database changes.

Funcion: User.record_failed_login

- * Ubicacion: app/models.py:61
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.
- * Efectos principales:
- * Persists database changes.

Funcion: User.reset_failed_login

- * Ubicacion: app/models.py:72
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

- * Efectos principales:
- * Persists database changes.

Funcion: Ticket.default_categories

- * Ubicacion: app/models.py:117
- * Tipo: Class method
- * Parametros: sin parametros declarados
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: Ticket.default_priorities

- * Ubicacion: app/models.py:121
- * Tipo: Class method
- * Parametros: sin parametros declarados
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: Ticket.categories

- * Ubicacion: app/models.py:125
- * Tipo: Class method
- * Parametros: sin parametros declarados
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: Ticket.priorities

- * Ubicacion: app/models.py:139
- * Tipo: Class method
- * Parametros: sin parametros declarados
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: Ticket.statuses

- * Ubicacion: app/models.py:153
- * Tipo: Class method
- * Parametros: sin parametros declarados
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: Ticket.sla_hours_by_priority

- * Ubicacion: app/models.py:162
- * Tipo: Class method
- * Parametros: priority
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: Ticket.is_closed

- * Ubicacion: app/models.py:171
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: Ticket.sla_warning_window_hours

- * Ubicacion: app/models.py:174
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: Ticket.is_overdue

- * Ubicacion: app/models.py:179
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: Ticket.is_due_soon

- * Ubicacion: app/models.py:182
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: ApiToken.is_valid

- * Ubicacion: app/models.py:279
- * Tipo: Class method
- * Parametros: self
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: AuditLog.log_action

- * Ubicacion: app/models.py:319
- * Tipo: Class method
- * Parametros: user_id, action, resource_type, resource_id, ip_address, user_agent, status, details
- * Que hace: Implements model/business behavior encapsulated in class logic.
- * Efectos principales:
- * Persists database changes.
- * Creates or stages new records in database session.

- * Rolls back transaction on error.

Funcion: PasswordHistory.check_password_reuse

- * Ubicacion: app/models.py:350
- * Tipo: Class method
- * Parametros: user_id, password, max_history
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: PasswordHistory.add_to_history

- * Ubicacion: app/models.py:364
- * Tipo: Class method
- * Parametros: user_id, password_hash, commit
- * Que hace: Implements model/business behavior encapsulated in class logic.
- * Efectos principales:
- * Persists database changes.
- * Creates or stages new records in database session.

Modulo: app/routes.py

Funcion: index

- * Ubicacion: app/routes.py:8
- * Tipo: HTTP endpoint
- * Ruta: /
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route / with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Efectos principales:
- * Renders HTML template response.

Funcion: health_check

- * Ubicacion: app/routes.py:13
- * Tipo: HTTP endpoint
- * Ruta: /health
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /health with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Efectos principales:

- * Returns JSON response.

Funcion: presence_ping

- * Ubicacion: app/routes.py:20
- * Tipo: HTTP endpoint
- * Ruta: /presence-ping
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /presence-ping with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.

Funcion: set_language

- * Ubicacion: app/routes.py:26
- * Tipo: HTTP endpoint
- * Ruta: /lang/
- * Metodos HTTP: GET
- * Parametros: lang
- * Que hace: Exposes route /lang/ with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Efectos principales:
- * Redirects browser to another route.

Funcion: about_us

- * Ubicacion: app/routes.py:34
- * Tipo: HTTP endpoint
- * Ruta: /quienes-somos
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /quienes-somos with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Efectos principales:
- * Renders HTML template response.

Funcion: terms_and_conditions

- * Ubicacion: app/routes.py:39
- * Tipo: HTTP endpoint
- * Ruta: /terminos-y-condiciones

- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /terminos-y-condiciones with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Efectos principales:
- * Renders HTML template response.

Modulo: app/schema_sync.py

Funcion: _ensure_columns

- * Ubicacion: app/schema_sync.py:46
- * Tipo: Internal helper
- * Parametros: table_name, columns
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.
- * Efectos principales:
- * Persists database changes.

Funcion: ensure_runtime_schema

- * Ubicacion: app/schema_sync.py:60
- * Tipo: Application helper
- * Parametros: sin parametros declarados
- * Que hace: Provides shared application utility behavior used by multiple flows.
- * Efectos principales:
- * Rolls back transaction on error.

Modulo: app/security.py

Funcion: PasswordValidator.validate

- * Ubicacion: app/security.py:47
- * Tipo: Class method
- * Parametros: password
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: TOTPManager.generate_secret

- * Ubicacion: app/security.py:88
- * Tipo: Class method
- * Parametros: sin parametros declarados
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: TOTPManager.generate_totp

- * Ubicacion: app/security.py:93
- * Tipo: Class method
- * Parametros: secret
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: TOTPManager.get_provisioning_uri

- * Ubicacion: app/security.py:98
- * Tipo: Class method
- * Parametros: secret, name, issuer
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: TOTPManager.get_qr_code_svg

- * Ubicacion: app/security.py:107
- * Tipo: Class method
- * Parametros: provisioning_uri
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: TOTPManager.verify_token

- * Ubicacion: app/security.py:128
- * Tipo: Class method
- * Parametros: secret, token
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: TOTPManager.generate_backup_codes

- * Ubicacion: app/security.py:138
- * Tipo: Class method
- * Parametros: count
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: TOTPManager.verify_backup_code

- * Ubicacion: app/security.py:144
- * Tipo: Class method
- * Parametros: codes_str, code
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: FileSecurityValidator.validate

- * Ubicacion: app/security.py:172

- * Tipo: Class method
- * Parametros: file_storage
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: FileSecurityValidator._detect_mime_type

- * Ubicacion: app/security.py:221
- * Tipo: Class method
- * Parametros: file_header, filename
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: FileSecurityValidator.scan_for_malware

- * Ubicacion: app/security.py:250
- * Tipo: Class method
- * Parametros: file_path
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: AuditHelper.get_client_ip

- * Ubicacion: app/security.py:285
- * Tipo: Class method
- * Parametros: sin parametros declarados
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: AuditHelper.get_user_agent

- * Ubicacion: app/security.py:292
- * Tipo: Class method
- * Parametros: sin parametros declarados
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: AuditHelper.log_login_attempt

- * Ubicacion: app/security.py:297
- * Tipo: Class method
- * Parametros: user, success, ip_address, user_agent
- * Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: AuditHelper.log_password_change

- * Ubicacion: app/security.py:317
- * Tipo: Class method
- * Parametros: user_id, ip_address, user_agent

* Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: AuditHelper.log_2fa_setup

* Ubicacion: app/security.py:336

* Tipo: Class method

* Parametros: user_id, enabled, ip_address, user_agent

* Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: AuditHelper.log_ticket_action

* Ubicacion: app/security.py:355

* Tipo: Class method

* Parametros: user_id, action, ticket_id, ip_address, user_agent

* Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: EncryptionManager.get_cipher

* Ubicacion: app/security.py:378

* Tipo: Class method

* Parametros: sin parametros declarados

* Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: EncryptionManager.encrypt

* Ubicacion: app/security.py:400

* Tipo: Class method

* Parametros: data

* Que hace: Implements model/business behavior encapsulated in class logic.

Funcion: EncryptionManager.decrypt

* Ubicacion: app/security.py:413

* Tipo: Class method

* Parametros: encrypted_data

* Que hace: Implements model/business behavior encapsulated in class logic.

Modulo: app/tickets.py

Funcion: utcnow

* Ubicacion: app/tickets.py:20

* Tipo: Application helper

* Parametros: sin parametros declarados

- * Que hace: Provides shared application utility behavior used by multiple flows.

Funcion: _t

- * Ubicacion: app/tickets.py:25
- * Tipo: Internal helper
- * Parametros: text
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.

Funcion: _ticket_read_access

- * Ubicacion: app/tickets.py:29
- * Tipo: Internal helper
- * Parametros: ticket
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.

Funcion: _ticket_chat_access

- * Ubicacion: app/tickets.py:33
- * Tipo: Internal helper
- * Parametros: ticket
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.

Funcion: _serialize_comment

- * Ubicacion: app/tickets.py:41
- * Tipo: Internal helper
- * Parametros: ticket, comment
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.

Funcion: _safe_datetime_text

- * Ubicacion: app/tickets.py:59
- * Tipo: Internal helper
- * Parametros: value, fmt
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.

Funcion: _ticket_timeline

- * Ubicacion: app/tickets.py:63
- * Tipo: Internal helper
- * Parametros: ticket
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.

Funcion: _reopen_deadline

- * Ubicacion: app/tickets.py:97
- * Tipo: Internal helper
- * Parametros: ticket
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.

Funcion: _can_reopen_ticket

- * Ubicacion: app/tickets.py:106
- * Tipo: Internal helper
- * Parametros: ticket
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.

Funcion: handle_connect

- * Ubicacion: app/tickets.py:114
- * Tipo: Socket event
- * Parametros: sin parametros declarados
- * Que hace: Handles a real-time Socket.IO event for chat/presence synchronization.

Funcion: handle_join_ticket_room

- * Ubicacion: app/tickets.py:152
- * Tipo: Socket event
- * Parametros: data
- * Que hace: Handles a real-time Socket.IO event for chat/presence synchronization.

Funcion: handle_disconnect

- * Ubicacion: app/tickets.py:174
- * Tipo: Socket event
- * Parametros: sin parametros declarados
- * Que hace: Handles a real-time Socket.IO event for chat/presence synchronization.

Funcion: handle_ticket_chat_message

- * Ubicacion: app/tickets.py:194
- * Tipo: Socket event
- * Parametros: data
- * Que hace: Handles a real-time Socket.IO event for chat/presence synchronization.
- * Efectos principales:
- * Persists database changes.
- * Creates or stages new records in database session.

Funcion: _attachments_dir

- * Ubicacion: app/tickets.py:213
- * Tipo: Internal helper
- * Parametros: sin parametros declarados
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.

Funcion: _save_attachment

- * Ubicacion: app/tickets.py:232
- * Tipo: Internal helper
- * Parametros: file_storage, ticket_id
- * Que hace: Supports internal workflow orchestration and is not intended as public endpoint.
- * Efectos principales:
- * Creates or stages new records in database session.

Funcion: list_tickets

- * Ubicacion: app/tickets.py:284
- * Tipo: HTTP endpoint
- * Ruta: /
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route / with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Efectos principales:
- * Renders HTML template response.
- * Sends user-visible feedback message.

Funcion: create_ticket

- * Ubicacion: app/tickets.py:450
- * Tipo: HTTP endpoint
- * Ruta: /create
- * Metodos HTTP: GET, POST
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /create with HTTP methods GET, POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Efectos principales:

- * Persists database changes.
- * Creates or stages new records in database session.
- * Renders HTML template response.
- * Sends user-visible feedback message.
- * Redirects browser to another route.

Funcion: kb_suggestions

- * Ubicacion: app/tickets.py:503
- * Tipo: HTTP endpoint
- * Ruta: /kb-suggestions
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /kb-suggestions with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Efectos principales:
- * Returns JSON response.

Funcion: request_password_reset_ticket

- * Ubicacion: app/tickets.py:525
- * Tipo: HTTP endpoint
- * Ruta: /request-password-reset
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /request-password-reset with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Efectos principales:
- * Persists database changes.
- * Creates or stages new records in database session.
- * Sends user-visible feedback message.
- * Redirects browser to another route.

Funcion: download_attachment

- * Ubicacion: app/tickets.py:581
- * Tipo: HTTP endpoint
- * Ruta: /attachments/download

- * Metodos HTTP: GET
- * Parametros: attachment_id
- * Que hace: Exposes route /attachments/download with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Efectos principales:
- * Loads resource or returns 404 automatically.
- * Can terminate request with HTTP error code.

Funcion: export_tickets

- * Ubicacion: app/tickets.py:598
- * Tipo: HTTP endpoint
- * Ruta: /export
- * Metodos HTTP: GET
- * Parametros: sin parametros declarados
- * Que hace: Exposes route /export with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.

Funcion: ticket_detail

- * Ubicacion: app/tickets.py:681
- * Tipo: HTTP endpoint
- * Ruta: /
- * Metodos HTTP: GET, POST
- * Parametros: ticket_id
- * Que hace: Exposes route / with HTTP methods GET, POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Efectos principales:
- * Persists database changes.
- * Creates or stages new records in database session.
- * Renders HTML template response.
- * Sends user-visible feedback message.
- * Loads resource or returns 404 automatically.
- * Broadcasts real-time event via Socket.IO.
- * Redirects browser to another route.

Funcion: reopen_ticket

- * Ubicacion: app/tickets.py:777
- * Tipo: HTTP endpoint
- * Ruta: //reopen
- * Metodos HTTP: POST
- * Parametros: ticket_id
- * Que hace: Exposes route //reopen with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Efectos principales:
- * Persists database changes.
- * Creates or stages new records in database session.
- * Sends user-visible feedback message.
- * Triggers email notification flow.
- * Loads resource or returns 404 automatically.
- * Can terminate request with HTTP error code.
- * Redirects browser to another route.

Funcion: submit_csat

- * Ubicacion: app/tickets.py:836
- * Tipo: HTTP endpoint
- * Ruta: //csat
- * Metodos HTTP: POST
- * Parametros: ticket_id
- * Que hace: Exposes route //csat with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Efectos principales:
- * Persists database changes.
- * Sends user-visible feedback message.
- * Loads resource or returns 404 automatically.
- * Can terminate request with HTTP error code.
- * Redirects browser to another route.

Funcion: ticket_chat_feed

- * Ubicacion: app/tickets.py:863

- * Tipo: HTTP endpoint
- * Ruta: //chat-feed
- * Metodos HTTP: GET
- * Parametros: ticket_id
- * Que hace: Exposes route //chat-feed with HTTP methods GET. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Efectos principales:
- * Returns JSON response.
- * Loads resource or returns 404 automatically.

Funcion: delete_ticket

- * Ubicacion: app/tickets.py:876
- * Tipo: HTTP endpoint
- * Ruta: //delete
- * Metodos HTTP: POST
- * Parametros: ticket_id
- * Que hace: Exposes route //delete with HTTP methods POST. Handles web request lifecycle and returns UI/API output.
- * Seguridad/Permisos:
- * Requires authenticated session.
- * Efectos principales:
- * Sends user-visible feedback message.
- * Loads resource or returns 404 automatically.
- * Redirects browser to another route.

5. Notas operativas

- * Revisar POST_LAUNCH_CHECKLIST.md para validacion post despliegue.
- * Para regenerar este manual, ejecutar tools/generate_detailed_manual.py y luego tools/build_manual_pdf.py.

Fin del documento.